

Glossary

The following is a compilation of some of the technical terms and climbing jargon used throughout this book.

active recovery—The restoration of homeostasis following vigorous exercise that involves continued light-intensity movement; facilitates faster recovery by enhancing lactate removal from the blood.

acute—Having rapid onset and severe symptoms.

adaptive response—Physiological changes in structure or function; particularly related to response to a training overload.

adipose tissue—Body fat.

aerobic energy system—ATP production via oxidation of glycogen, fats, and, occasionally, proteins.

agonist—A muscle directly engaged in a muscular contraction.

anaerobic glycolysis—ATP production in the muscles involving the breakdown of glycogen in the absence of oxygen; metabolic by-products, including chiefly hydrogen ions, result in rapid fatigue and cessation of physical activity.

antagonist—A muscle providing an opposing force to the primary muscles of action (agonist).

ape index—Fingertip-to-fingertip distance (across your chest with arms out to each side) minus your height; a positive ape index is associated with above-average reach for a given height.

arousal—An internal state of alertness or excitement.

artery—A vessel that carries oxygenated blood away from the heart to the tissues of the body.

arthritis—A disease that causes inflammation, swelling, and pain in the joints.

ATP—Adenosine triphosphate; a high-energy molecule that is stored in the muscles in very small amounts; it's the body's ultimate fuel source.

atrophy—Gradual shrinking and deconditioning of muscle tissue from disuse.

backstepping—Outside edging on a foothold that is behind you while climbing a move with your side to the wall.

barndoor—Sideways swinging or uncontrolled turning of the body resulting from poor balance or body positioning.

basal metabolic rate—The minimum level of energy required to sustain the body's vital functions.

beta—Any prior information about a route, including sequence, rests, gear, clips, and so on.

biological value (BV)—A method for evaluating protein sources; a high BV protein source has a high percentage of nutrients actually absorbed from the human intestine as opposed to excreted.

blocked practice—A practice routine in which a specific task is practiced repeatedly, as in working a crux move or sequence.

campus (or campusing)—Climbing an overhanging section of rock or artificial wall with no feet, usually in a dynamic left-hand, right-hand, left-hand (and so forth) sequence.

campus training—A climbing-specific form of plyometric exercise, developed by Wolfgang Güllich at the Campus Center, a weight-lifting facility at the University of Nürnberg, Germany.

capillary—The tiny blood vessels that receive blood flow from the arteries, interchange substances between the blood and the tissues, and return the blood to the veins.

capillary density—The number of capillaries per unit area of muscle tissue. Capillary density increases, mainly in slow-twitch fibers, in response to aerobic training.

catabolic—A breaking-down process in the body, as in muscle breakdown during intense exercise.

center of mass—The theoretical point on which the total effect of gravity acts on the body.

chronic—Continuing over time.

concentric contraction—Any movement involving a shortening of muscle fibers while developing tension, as in the biceps muscle during a pull-up.

contact strength—Initial grip strength upon touching a handhold; directly related to the rate of force development in the finger flexor muscles.

cortisol—A hormone, released in response to emotional or exercise stress, that promotes fat utilization, inhibits inflammatory response, and facilitates breakdown of muscle proteins for energy.

cortisone—A synthetic form of cortisol used (injected) as an anti-inflammatory.

creatine phosphate (CP)—A high-energy phosphate compound stored in skeletal muscle and used to supply energy for brief, high-intensity muscle action.

crimp grip—The most natural (and stressful) way to grip a rock hold, characterized by hyperextension of the first joint in the fingers and nearly full flexion of the second joint.

crux—The hardest move, or sequence of moves, on a route.

deadpoint—The high position in a dynamic movement where, for a moment, all motion stops.

detraining—Reversal of positive adaptations to chronic exercise upon cessation of an exercise program.

drop-knee—An exaggerated backstep, commonly used on overhanging rock, where the inside knee is dropped toward the ground, resulting in a stable chimney-like position.

dynamic move—An explosive leap for a hold otherwise out of reach.

dyno—Short for “dynamic.”

eccentric contraction—A muscle action in which the muscle resists as it is forced to lengthen, as in the biceps during the lowering phase of a pull-up.

electrolyte—A substance that, in solution, is capable of conducting electricity. Certain electrolytes are essential to the electrochemical functioning of the body.

endurance—The ability to perform physical work for an extended period of time. Cardiovascular endurance is directly related to VO_2 max, whereas muscular endurance is influenced by circulation and available oxygen and energy substrate.

enzyme—A protein molecule that aids chemical reactions.

epicondylitis—Inflammation of the tendon origins of the forearm flexors (medial) or extensors (lateral) near the elbow.

ergogenic—Performance enhancing.

estrogen—The sex hormone that predominates in females, but also has some functions in males.

extension—A movement that moves the two ends of a jointed body part away from each other, as in straightening the arm.

fast-twitch fibers—The muscle fiber type that contracts quickly and is used mostly during intense, powerful movements.

flagging—A climbing technique in which one foot is crossed behind the other to avoid barn-dooring and to improve balance.

flash—To climb a route on the first try without ever having touched it, but with the aid of beta.

flash pump—A metaboreflex characterized by a rapid, often vicious, muscular pump resulting from strenuous training or climbing without first performing a proper, gradual warm-up.

flexion—A movement that brings the ends of a body part closer together, as in bending the arm.

G-Tox—A technique that uses gravity to help speed recovery from a forearm pump. It involves alternating, every five to ten seconds, the position of the resting arm between the normal hanging-at-your-side position and a raised-hand position above your shoulder.

glycemic index (GI)—A scale that classifies how the ingestion of various foods affects blood sugar levels in comparison with the ingestion of straight glucose.

glycogen—Compound chains of glucose stored in the muscle and liver for use during aerobic or anaerobic exercise.

Golgi tendon organ—Sensory receptors located between the muscle and its tendon that are sensitive to the stretch of the muscle tendon produced during muscular contraction.

gripped—Extremely scared.

hangdogging—Climbing a route, usually bolt-to-bolt, with the aid of a rope to hang and rest while practicing the sequence.

heel hook—Use of the heel on a hold, usually near chest level, to aid in pulling and balance.

homeostasis—The body's tendency to maintain a steady state despite external changes.

honed—In extremely good physical shape; with low body fat.

hormone—A chemical secreted into the bloodstream to regulate the function of a certain organ.

hyperemia—Increased blood flow in the working muscles during exercise or as a result of deep sports massage.

Hypergravity Isolation Training (HIT)—A highly refined and specific method of training maximum finger strength and upper-body power by climbing on identical finger holds (isolation) with greater than body weight (hypergravity). Also known as Hörst Isolation Training.

hypertrophy—Enlargement in size (e.g., muscular hypertrophy).

insertion—The point of attachment of a muscle to a distal or relatively more movable bone.

insulin—A hormone that decreases the blood glucose level by driving glucose from the blood into muscle and fat cells.

interval training—A method of anaerobic endurance training that involves brief periods of intense training interspaced with periods of rest or low-intensity training.

isometric—A muscular contraction resulting in no shortening movement of the muscle.

kinesthetic—The sense derived from muscular contractions and limb movements. See proprioception.

killer—Extraordinarily good, as in a killer route.

lean body weight—The weight of the body, less the weight of its fat.

ligament—Fibrous tissue that connects bone to bone, or bone to cartilage, to hold together and support the joints.

lunge—An out-of-control dynamic move; an explosive jump for a far-off hold.

macronutrients—Basic nutrients needed for energy, cell growth, and organ function (carbohydrates, fat, and protein).

maximum strength—The peak force of a muscular contraction, irrespective of the time element. Also known as "limit strength."

micronutrients—Noncaloric nutrients needed in very small amounts, such as vitamins and minerals.

modeling—A learning technique where an individual watches, then attempts, a skill as performed properly by another person.

motor learning—A set of internal processes associated with practice or experience leading to a relatively permanent gain in performance capability.

motor program—Motor programs are the way in which the brain controls movements. Motor programs become more hardwired and lasting via deliberate practice.

motor skill—A skill where the primary determinant of success is the movement component itself.

motor unit—A motor neuron, together with a group of muscle cells, stimulated in an all-or-nothing response.

NSAID—Non-steroidal anti-inflammatory drugs, usually available over the counter, that reduce pain, fever, and inflammation.

on-sight—When a route is climbed on the first try and with absolutely no prior information of any kind.

open-hand grip—The less-stressful finger grip involving only slight flexion of the finger joints.

osteoarthritis—A joint disease of older persons in which cartilage in the joints wears down and there is bone growth at the edges of the joint.

overload—Subjecting a part of the body to greater efforts (intensity or volume) than it is accustomed to in order to elicit a training response.

overtraining—Constant severe training that does not provide adequate time for recovery; symptoms include increased frequency of injury, decreased performance, irritability, and apathy.

overuse—Excessive repeated exertion or shock that results in injuries such as inflammation of the muscles and tendons.

plyometric—An exercise that suddenly preloads and forces the stretching of a muscle an instant prior to its concentric contraction, as in dynamic up-and-down campus training. Also known as reactive or shock training.

power—A measure of both force and speed (speed = distance x time) of a muscular contraction through a given range of motion. Power is the explosive aspect of strength.

pronation—The inward turning of a body part, as in turning the forearm inward and the palm facedown.

proprioception—Sensory information from within the body arising from the sensory receptors found in muscles, tendons, joints, and the inner ear that detect the motion or position of the limbs and body.

proprioceptors—Sensory receptors found in muscles, tendons, joints, and the inner ear that detect the motion or position of the limbs and body, thus providing kinesthetic awareness.

psyched—Raring to go or very happy.

pumped—When the muscles become engorged with blood due to extended physical exertion.

random practice—A practice sequence in which tasks from several classes are experienced in random order over consecutive trials.

reactive training—A power-building exercise that couples, in rapid succession, a forceful eccentric contraction with an explosive concentric contraction.

recommended dietary allowances

(RDA)—Quantities of specific vitamins, minerals, and other nutrients needed daily that have been judged adequate for maintenance of good nutrition. Developed by the Food and Nutrition Board of the National Academy of Science.

recruitment—The systematic increase in the number of active motor units called upon during muscular contraction.

redpoint—Lead climbing a route bottom-to-top in one push.

schema—A set of rules, usually developed and applied unconsciously by the motor system in the brain and spinal cord, relating how to move and adjust muscle forces, body positions, and so forth, given the parameters at hand, such as steepness of the rock, friction qualities, holds being used, and type of terrain.

send—short for “ascend,” as in *I’m going to send that climb . . .* or, past tense, *I sent my project!*

sharp end—The lead climber's end of the rope.

skill—The capability to bring about an end result with maximum certainty, minimum energy, and minimum time.

slow-twitch fibers—The muscle fiber type that contracts slowly and is used most in moderate-intensity endurance activities such as easy to moderate climbing or running.

sport climbing—Usually refers to any indoor or outdoor climbing on bolt-protected routes.

spotter—A person designated to slow the fall of a boulderer, with the main goal of keeping the boulderer's head from hitting the ground.

stabilizer muscle—A muscle that is stimulated to help anchor or stabilize the position of a bone.

strength—The amount of muscle force that can be exerted; speed and distance are not factors of strength.

supination—Rotation of the forearm outward and palm-upward.

synovial fluid—A viscid fluid secreted by the membrane lining joints, tendon sheaths, and bursae to lubricate and cushion them during movement.

tendinitis—An acute disorder involving the inflammation of a tendon and synovial membrane at a joint.

tendinopathy—A general term that just refers to tendon injury, without specifying a particular type of injury such as tendinitis (inflammation) or tendinosis (failed healing).

tendinosis—Chronic tendon pain due to an accumulation of microscopic injuries that don't heal properly; the main problem, then, is failed healing, not inflammation.

tendon—A white fibrous cord of dense connective tissue that attaches muscle to bone.

trad—Short for a traditional climb (or climber) that requires natural protection placements.

training effect—A basic principle of exercise science that states that adaptation occurs from an exercise only in those parts or systems of the body that are stressed by the exercise.

transfer of learning—The gain or loss in proficiency on one task as a result of practice or experience on another task.

trigger point—A long-lasting muscle spasm or contracture often associated with local muscular fatigue or injury.

tweak—To injure, as in a tweaked finger tendon.

variable practice—Practice in which many variations on a class of actions are performed; opposite of blocked practice.

vein—A vessel that returns blood from the various parts of the body to the heart.

visualization—Controlled and directed imagery that can be used for awareness building, monitoring and self-regulation, healing, and, most important, as mental programming for good performances.

VO₂ max—Maximal oxygen uptake, as in the measurement of maximum aerobic power.

wired—Known well, as in a wired route.

working—Practicing the moves on a difficult route via top rope or hangdogging.

A full-page photograph of a young man with curly brown hair, wearing a blue t-shirt and dark shorts, climbing a steep, light-colored rock face. He is positioned vertically, with his head near the top of the frame and his feet lower down. He is wearing a climbing harness and yellow shoes. His right arm is extended upwards, reaching for a hold, while his left arm is bent. A climbing rope is visible, attached to his harness and running down the rock face. The background shows a vast, hazy landscape with green trees and a distant horizon under a cloudy sky. The text is located in the upper right corner of the image.

*Twelve-year-old Jonathan
Hörst sending Cow Reggae
(5.13b), Wild Iris, Wyoming.*
SAVANNAH CUMMINS

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